Facultat d'Informàtica de Barcelona

OPTICAL MUSIC RECOGNITION FOR JAZZ LEAD SHEETS: A CRNN-CTC APPROACH FOR MONOPHONIC LINES

POL CASANOVAS PUIG

Thesis supervisor
MANEL FRIGOLA BOURLON (Department of Automatic Control)

GEP tutor
XAVIER MORALES SOROLLA

Degree
Bachelor's Degree in Informatics Engineering (Computing)

GEP Deliverable 4: Final Document

Facultat d'Informàtica de Barcelona (FIB)

Universitat Politècnica de Catalunya (UPC) – BarcelonaTech

March 2026

Contents

1 Introduction and Context

1.1 Academic and Professional Framework

This project is developed in the academic context of the Bachelor's Thesis (TFG) at FIB–UPC, under modality A, as part of the requirements to complete the Computer Science degree at the Universitat Politècnica de Catalunya (UPC). The project is supervised by Manel Frigola Bourlon and guided in project management by Xavier Morales Sorolla.

The work is framed within the Computacio specialty and focuses on the intersection of music and artificial intelligence. The topic is motivated by a practical need in music education and performance: converting printed lead sheets into machine-readable formats with less manual effort.

1.2 Domain Terms and Concepts

The project uses the following key concepts:

- **Optical Music Recognition (OMR):** automatic conversion of sheet-music images into symbolic formats such as MusicXML or MIDI [1].
- **CRNN-CTC:** architecture combining convolutional feature extraction with recurrent sequence modeling and CTC alignment.
- **Lead sheet:** compact score representation used in jazz, typically containing melody and harmonic symbols.
- **Monophonic transcription:** recognition of a single melodic line, without simultaneous note voices.
- **SER (Symbol Error Rate):** sequence-level metric used to evaluate transcription quality [2].

1.3 Problem Statement

Jazz lead sheets are often available only as scanned or printed documents. Manual transcription into notation software is slow, repetitive, and prone to mistakes. Generic OMR systems are usually optimized for classical full-score notation and often perform poorly on lead-sheet conventions (compact layout, jazz engraving style, mixed symbolic elements).

The project addresses this gap by developing a reproducible end-to-end prototype tailored to monophonic lead-sheet lines, prioritizing practical feasibility within the constraints of a Bachelor's thesis.

1.4 Stakeholders

- **Musicians and students:** need searchable, editable, and transposable scores.
- **Educators:** require digital materials for classes and exercises.
- **Archivists and libraries:** benefit from improved preservation and retrieval of music documents.
- **MIR and OMR researchers:** can reuse dataset, code, and evaluation outputs.

2 Justification of the Chosen Alternative

2.1 Existing Alternatives

Current OMR systems can be grouped into three families [1]:

1. **Traditional modular pipelines** (staff detection/removal, segmentation, classification), with known error propagation issues [3].
2. **CRNN-CTC sequence models**, well suited to left-to-right staff-line transcription and widely used in modern monophonic OMR benchmarks [4], [5].
3. **Transformer-based end-to-end models**, stronger for page-level and polyphonic settings but heavier in data and compute needs [5], [6], [7].

2.2 Selection Criteria

The decision is based on: expected accuracy in monophonic lines, implementation complexity, compatibility with available GPU resources, reproducibility, and suitability for an incremental development plan.

Table 1: Comparison of alternatives against project criteria (++, +, -, −).

Criterion	**Traditional pipeline**	**CRNN-CTC**	**Transformer**
Monophonic line accuracy	-	+	++
Computational feasibility	++	+	−
Data requirements	++	+	−
Implementation complexity	+	+	−
Reproducibility in TFG timeline	+	++	-
Extension potential (future work)	-	+	++

This comparison shows why CRNN-CTC is the best fit for this TFG stage: it provides enough modeling capacity for monophonic lead sheets while remaining trainable and debuggable within the available timeframe and hardware.

2.3 Chosen Alternative

The selected solution is a **CRNN-CTC architecture** trained for **monophonic line transcription**. This option provides a good compromise between performance and feasibility: it is lighter than transformer alternatives, has strong prior art in OMR datasets, and supports robust sequence prediction without requiring symbol-level segmentation.

The decision is also justified by risk control. A modular baseline can be implemented early and used as a lower bound, while the CRNN-CTC model can be iteratively improved through data and training refinements. This creates a robust path where the project still produces valid results even if advanced extensions cannot be completed in time.

2.4 Methodology and Rigour

The project follows an **iterative and incremental** methodology, with sprint-based checkpoints and supervisor reviews. Each cycle produces measurable outputs (datasets, training logs, SER values, and error analyses), enabling evidence-based replanning.

Monitoring tools include Git/GitHub for version control, experiment logs for model runs, and periodic milestone meetings with the supervisor to validate priorities and risk responses.

Rigour is enforced through four concrete practices:

- **Traceability:** every milestone output is linked to repository commits and documented decisions.
- **Reproducibility:** data generation, training, and evaluation are script-driven, with fixed configuration files.
- **Quantitative control:** SER is measured at each significant training iteration, with side metrics for failure analysis.
- **Corrective loop:** each sprint ends with a review that can adjust scope, estimates, and risk responses.

3 Project Scope

3.1 General Objective

Develop and evaluate a proof-of-concept OMR system based on CRNN-CTC capable of transcribing monophonic jazz lead-sheet lines into a machine-readable symbolic representation.

3.2 Specific Objectives

- Review state-of-the-art OMR methods and identify practical design decisions.
- Build a baseline pipeline and a CRNN-CTC model for comparison.
- Construct a suitable training/evaluation dataset using synthetic rendering and augmentation.
- Evaluate model quality with SER and qualitative error analysis.
- Release a reproducible code and configuration workflow.

3.3 Functional Requirements

- Input: scanned or photographed score images.
- Processing: extraction of monophonic line images and sequence recognition.
- Output: symbolic sequence preserving pitch, duration, rests, and meter context.
- Evaluation: quantitative metrics and comparable baseline results.

3.4 Non-Functional Requirements

- **Accuracy:** improve over the classical baseline in SER.
- **Reproducibility:** scripts and configuration available in repository.
- **Feasibility:** training on consumer-grade GPU (RTX 3060 12GB class).
- **Modularity:** separable stages for data, model, and evaluation.

3.5 Scope Boundaries

In scope: monophonic lead-sheet lines, model training and evaluation, synthetic data pipeline, and comparative analysis.

Out of scope: full polyphonic transcription, complete page-layout understanding, handwritten notation coverage, and large-scale transformer training from scratch.

3.6 Main Risks and Obstacles

- Limited target-domain labels.
- Error propagation from preprocessing into recognition.
- CTC alignment instability in early runs.
- GPU-time constraints during iterative retraining.
- Scope creep toward polyphony before reaching monophonic targets.

4 Project Planning

4.1 Task-Level Planning, Dependencies, and Resources

The planning spans 21 weeks (February–June 2026), organized into 5 technical sprints plus a transversal management package (WP0). Table 2 details tasks at task level, estimated hours, dependencies, and required resources.

The precedence logic is the following: Sprint 1 provides the analytical and baseline foundation; Sprint 2 builds the dataset and label encoding required by training; Sprint 3 develops the first CRNN-CTC model; Sprint 4 performs the iterative train-evaluate-improve loop; Sprint 5 consolidates final evaluation and thesis delivery. WP0 runs in parallel to guarantee management continuity and documentation quality.

4.2 Gantt and Time Estimates

The schedule includes realistic task concurrence, review milestones, and a conditional extension buffer in Sprint 5. Figure 1 summarizes the full timeline.

Estimates are expressed in hours per task and were validated against available weekly workload (4–5 effective hours/day). Concurrency is intentionally limited to avoid unrealistic multitasking in a single-person project; overlap is only introduced when tasks are naturally compatible (for example, evaluation framework development while the first training run is ongoing).

Table 2: Task summary with estimates, dependencies, and resources.

Code	Task	h	Dependencies	Resources
WP0 – Project Management (130 h)				
T0.1	Context and scope definition	20	–	Zotero
T0.2	Temporal planning	12	T0.1	–
T0.3	Budget and sustainability	15	T0.2	–
T0.4	Final GEP integration	8	T0.3	–
T0.5	Supervisor meetings	25	Concurrent	–
T0.6	Thesis writing	50	Concurrent	LaTeX
WP1 – Sprint 1: Literature Review and Baseline (78 h)				
T1.1	OMR literature survey	25	–	Library access
T1.2	Dataset exploration	15	T1.1 (partial)	–
T1.3	Classical CV baseline implementation	25	T1.2	OpenCV
T1.4	Baseline evaluation	10	T1.3	–
T1.5	Sprint 1 review	3	T1.4	–
WP2 – Sprint 2: Dataset Construction (73 h)				
T2.1	Output encoding design	12	T1.5	music21
T2.2	Synthetic rendering pipeline	35	T2.1	LilyPond
T2.3	Data augmentation	15	T2.2	OpenCV
T2.4	Validation and splits	8	T2.3	–
T2.5	Sprint 2 review	3	T2.4	–
WP3 – Sprint 3: CRNN-CTC Development (75 h)				
T3.1	CRNN architecture design	25	T2.5	Consumer GPU
T3.2	Training pipeline implementation	20	T3.1	–
T3.3	Initial training run	15	T3.2	GPU
T3.4	Evaluation framework	12	T3.2 (partial)	–
T3.5	Sprint 3 review	3	T3.3, T3.4	–
WP4 – Sprint 4: Iterative Refinement (68 h)				
T4.1	Error analysis	10	T3.5	–
T4.2	Data pipeline improvements	15	T4.1	–
T4.3	Model refinement runs	25	T4.2	GPU
T4.4	Ablation studies	15	T4.3	GPU
T4.5	Sprint 4 review	3	T4.4	–
WP5 – Sprint 5: Final Evaluation and Defence (85 h)				
T5.1	Final evaluation	15	T4.5	GPU
T5.2	Qualitative error analysis	10	T5.1	–
T5.3	Extension exploration (conditional)	20	T5.1	GPU
T5.4	Final thesis writing	30	T5.2	LaTeX
T5.5	Defence preparation	10	T5.4	–
	TOTAL	**509**		

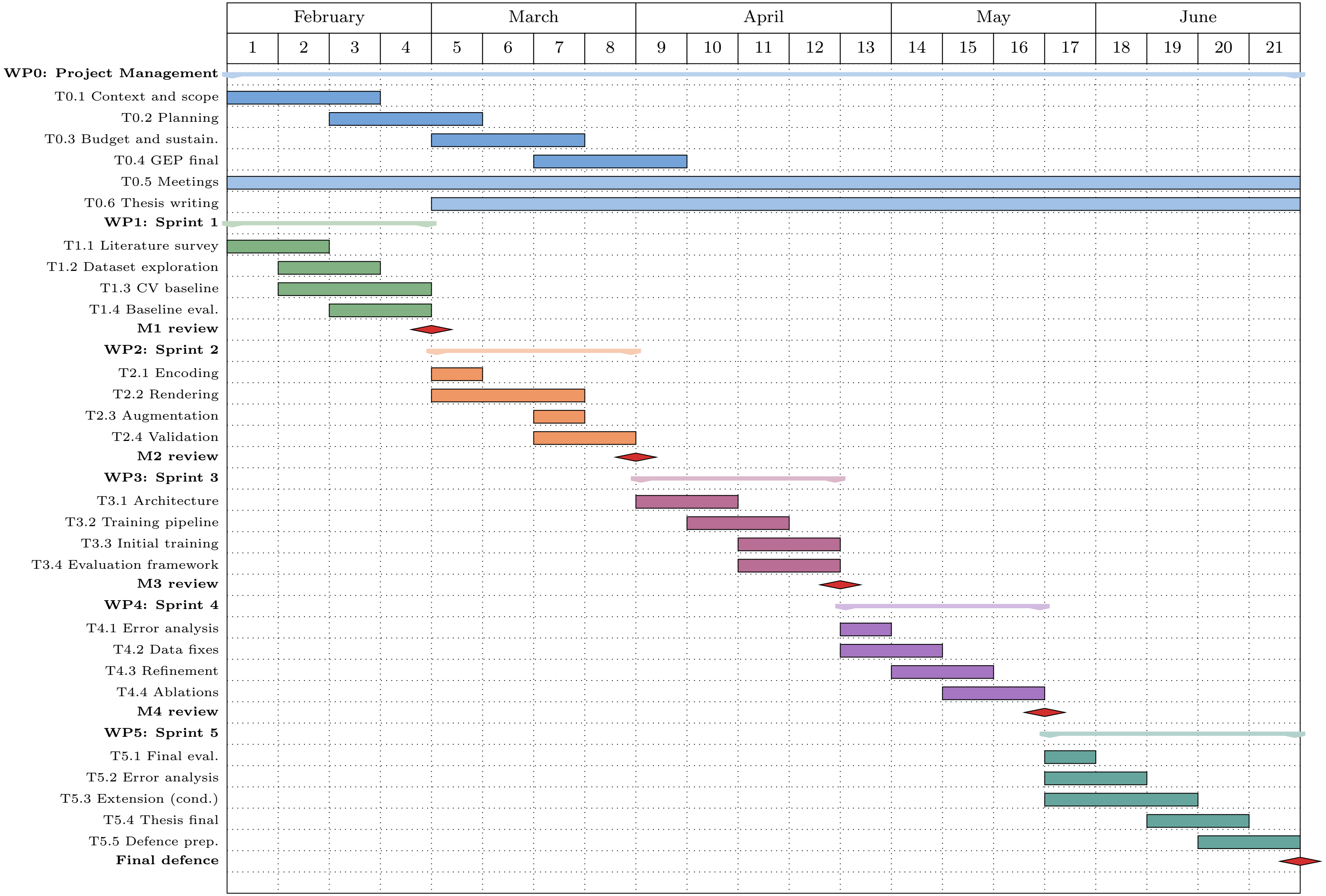

Figure 1: Gantt chart for weeks 1–21 (February–June 2026).

4.3 Alternative Plans and Risk Response

Table 3 identifies the main risks and explicit response plans.

Table 3: Risk register with alternative plans and estimated impact.

Risk	**L/I**	**Alternative plan**	**Duration impact**	**Extra resources**
Limited ground-truth data	H/H	Fall back to PrIMuS-only data and strengthen augmentation.	Reallocate up to 10 h.	None.
Staff detection instability	M/M	Use manually cropped line regions for critical evaluations.	Shift hours from T5.3 to T5.4.	None.
CTC underfitting/misalignment	M/H	Simplify vocabulary, retrain with cleaned subsets.	Up to 10 h from buffer.	Additional GPU time.
Compute constraints	L-M/M	Reduce model/data size or rent cloud GPU.	Up to 5 h setup.	Cloud GPU (€20–50).
Scope creep	M/H	Restrict extension to feasibility discussion only.	Recover up to 18 h.	None.

To improve schedule robustness, the plan includes explicit slack in three places: conservative estimation in high-uncertainty tasks (dataset rendering and model refinement), a conditional extension task that can be reduced without affecting core objectives, and progressive thesis writing during execution instead of postponing writing to the final weeks.

5 Economic Management

5.1 Budget Structure and Cost Identification

The budget follows the GEP framework and includes: human resources by activity, general costs, amortization, contingency, and incidentals [8].

5.2 Human Resources Cost Estimates

Hourly rates are based on Spanish ICT salary data and loaded with a 1.35 multiplier to account for employer social-security costs [9], [10].

Table 4: Roles and loaded hourly rates.

Role	Gross (€/year)	Loaded (€/year)	€/h
Project Manager	38,000	51,300	29.15
Research Analyst	33,000	44,550	25.31
ML Engineer	36,000	48,600	27.61
Technical Writer	30,000	40,500	23.01

In addition to work-package totals, Table 5 itemizes staff cost by activity, aligned with the Gantt tasks. A blended loaded rate of €26.75/h is used for planning-level allocation (rounded); this yields €13,615.57 total human resources cost.

Table 5: Staff cost by activity (task-level itemization).

Task	Description	Hours	Staff cost (€)
T0.1	Context and scope definition	20	535.00
T0.2	Temporal planning	12	321.00
T0.3	Budget and sustainability	15	401.25
T0.4	Final GEP integration	8	214.00
T0.5	Supervisor meetings	25	668.75
T0.6	Thesis writing	50	1,337.50
T1.1	OMR literature survey	25	668.75
T1.2	Dataset exploration	15	401.25
T1.3	Classical CV baseline implementation	25	668.75
T1.4	Baseline evaluation	10	267.50
T1.5	Sprint 1 review	3	80.25
T2.1	Output encoding design	12	321.00
T2.2	Synthetic rendering pipeline	35	936.25
T2.3	Data augmentation	15	401.25
T2.4	Validation and splits	8	214.00
T2.5	Sprint 2 review	3	80.25
T3.1	CRNN architecture design	25	668.75
T3.2	Training pipeline implementation	20	535.00
T3.3	Initial training run	15	401.25
T3.4	Evaluation framework	12	321.00
T3.5	Sprint 3 review	3	80.25
T4.1	Error analysis	10	267.50
T4.2	Data pipeline improvements	15	401.25
T4.3	Model refinement runs	25	668.75
T4.4	Ablation studies	15	401.25
T4.5	Sprint 4 review	3	80.25
T5.1	Final evaluation	15	401.25
T5.2	Qualitative error analysis	10	267.50
T5.3	Extension exploration (conditional)	20	535.00
T5.4	Final thesis writing	30	802.50
T5.5	Defence preparation	10	267.50
	TOTAL	**509**	**13,615.57**

Table 6: Human resources budget by work package.

WP	Work package	Hours	Cost (€)
WP0	Project management	130	3,482.50
WP1	Literature review and baseline	78	2,066.20
WP2	Dataset construction	73	1,992.55
WP3	CRNN-CTC development	75	2,075.37
WP4	Iterative refinement	68	1,859.10
WP5	Final evaluation and defence	85	2,139.85
	TOTAL HUMAN RESOURCES	**509**	**13,615.57**

5.3 General Costs, Amortization, and Incidentals

Table 7: General costs summary.

Item	Cost (€)
Hardware amortization (desktop, monitor, peripherals)	125.69
Software licenses (open-source stack)	0.00
Overheads (electricity + internet)	125.00
TOTAL GENERAL COSTS	**250.69**

Table 8: Incidentals linked to risk materialization scenarios.

Risk	Contingency action	Full cost (€)	Prob.	Weighted (€)
R1	Additional data cleaning and augmentation cycle	276.10	25%	69.03
R3	Retraining cycle with simplified vocabulary and tuning	276.10	20%	55.22
R4	Temporary cloud GPU rental and setup	178.05	15%	26.71
R5	Scope reduction in extension task (already internalized)	0.00	30%	0.00
			TOTAL INCIDENTALS	**150.96**

A contingency reserve of 15% is applied over human resources + general costs:

$$\text{Contingency} = 0.15 \times (13,615.57 + 250.69) = e2,079.94$$

Incidentals linked to the main risk responses are estimated at €150.96.

5.4 Budget Management Control

Budget tracking is performed at each sprint review with earned-value-inspired indicators:

- **Cost Variance (CV)**: CV = BCWP − ACWP.
- **Schedule Variance (SV)**: SV = BCWP − BCWS.
- **Cost Performance Index (CPI)**: CPI = BCWP/ACWP.

Corrective actions are triggered when: $|\text{CV}| > 10\%$ of sprint budget, $\text{SV} < -5\%$, or $\text{CPI} < 0.90$ for two consecutive reviews.

Table 9: Total budget.

Budget component	Cost (€)
Human resources	13,615.57
General costs	250.69
Contingency (15%)	2,079.94
Incidentals	150.96
TOTAL PROJECT BUDGET	**16,097.16**

6 Sustainability and Social Commitment

6.1 Self-Assessment Summary

After completing the sustainability self-assessment questionnaire, I identified that my strongest area is the economic dimension of sustainability in software projects, especially cost estimation, trade-off analysis, and management control. I am comfortable reasoning about whether a technical decision is financially realistic within a constrained project and about how planning choices affect viability and long-term maintenance cost. The questionnaire also showed a medium level in social reflection: I generally consider accessibility and user benefit, but I still tend to prioritize technical performance first and social impact second when under time pressure.

My weakest area is the environmental lifecycle perspective. Before this project, I mainly associated environmental impact with direct energy consumption during model training. The reflection process made me consider additional factors that I had not analyzed in depth: hardware manufacturing footprint, end-of-life device management, and the difference between one-time experimentation and sustained deployment impact. I also realized that selecting lighter architectures is not only a computational decision, but also an environmental one.

As a result, this self-assessment produced three concrete improvement commitments for the rest of the thesis: first, to quantify energy-related impact whenever possible instead of using only qualitative statements; second, to justify architecture choices with both performance and efficiency criteria; and third, to explicitly discuss who benefits from the resulting tool and under what limitations. Overall, the questionnaire was useful because it transformed sustainability from a formal reporting requirement into a design constraint that can guide technical decisions throughout the project lifecycle.

This reflection also improves my personal competency for future professional work: I now see project quality as a balance between technical success, economic feasibility, environmental responsibility, and social value.

6.2 Sustainability Matrix (Initial Milestone)

Table 10 summarizes the reflection using the economic, environmental, and social dimensions requested in the GEP sustainability guide [11].

6.2.1 Economic Dimension Reflection

PPP. The project cost is estimated at €16,097.16, dominated by specialized engineering time. This is coherent with a research-intensive software project where tooling and hardware costs are relatively low compared with human capital.

Useful life and state of the art. Current alternatives for lead-sheet digitization are manual transcription or generic OMR tools. Manual transcription is expensive in recurring cost and turnaround time. Generic OMR tools may require substantial correction for jazz notation specifics. A specialized and open-source CRNN-CTC model can lower recurrent cost by reducing correction workload and eliminating license barriers for many users.

Improvement contribution. The main economic improvement is not zero-cost digitization but a lower cost per corrected page after deployment, plus reproducibility benefits for research and education.

6.2.2 Environmental Dimension Reflection

PPP. The estimated project energy consumption is approximately 57.5 kWh, corresponding to around 13.2 kg CO_2 with the Spanish electricity factor used in this report [12], [13]. The largest contributor is GPU training.

Minimization plan. To reduce impact, the project uses a lightweight backbone, mixed-precision training, early stopping, and reuse of existing hardware. Data generation and training are organized in batches to avoid repeated redundant runs.

Useful life and improvement contribution. Once trained, inference has low marginal energy cost compared with retraining. Sharing model weights and scripts allows re-use of the one-time training cost across multiple users and academic iterations.

6.2.3 Social Dimension Reflection

PPP. In personal development terms, the project strengthens competencies in machine learning engineering, project planning, and scientific communication. It also reinforces ethical awareness on limitations and failure modes.

Useful life and real need. There is a real demand for digitizing lead sheets used in education and performance contexts. Beneficiaries include musicians, teachers, archivists, and researchers. The project can improve quality of life by reducing repetitive manual work, enabling quicker access to learning material, and facilitating preservation of music content.

Boundaries. Social value depends on transparent communication of current limitations (monophonic focus, expected error profile), avoiding overpromising in scenarios outside the validated scope.

Table 10: Sustainability matrix for the project.

Dimension	**PPP (design and development)**	**Useful life (deployment and impact)**
Economic	Project budget is estimated at €16,097.16, with explicit contingency and incidentals. Main cost driver is specialized engineering time.	Open-source release and fast local inference can reduce recurrent transcription costs compared with manual copying and commercial tools.
Environmental	Estimated energy use is approximately 57.5 kWh for the project, with mitigation via mixed precision, early stopping, and reuse of existing hardware [12], [13].	One-time training cost is amortized over repeated inference; local execution can reduce cloud dependency for common use cases.
Social	The project strengthens personal technical competencies and promotes responsible data and software practices.	Potential beneficiaries include musicians, educators, archivists, and researchers through improved accessibility, searchability, and preservation of music documents.

6.3 Commitment and Ethical Position

The project is aligned with social commitment through open-science practices (reproducible code, transparent metrics, and documented limitations). It does not process personal data and does not involve safety-critical automated decision making. Ethical limitations are explicitly documented, especially regarding performance boundaries outside monophonic lead-sheet scenarios.

References

[1] E. Shatri and G. Fazekas, "OPTICAL MUSIC RECOGNITION: STATE OF THE ART AND MAJOR CHALLENGES,"

[2] "Symbol Error Rate - an overview | ScienceDirect Topics," Accessed: Feb. 18, 2026. [Online]. Available: https://www.sciencedirect.com/topics/computer-science/symbol-error-rate

[3] C. Dalitz, M. Droettboom, B. Pranzas, and I. Fujinaga, "A Comparative Study of Staff Removal Algorithms," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 30, no. 5, pp. 753–766, May 2008, ISSN: 0162-8828. DOI: 10.1109/TPAMI.2007.70749 Accessed: Feb. 18, 2026. [Online]. Available: http://ieeexplore.ieee.org/document/4359356/

[4] "PrIMuS dataset," Accessed: Feb. 18, 2026. [Online]. Available: https://grfia.dlsi.ua.es/primus/

[5] F. Simonetta, R. Mondal, L. A. Ludovico, and S. Ntalampiras, "Optical Music Recognition in Manuscripts from the Ricordi Archive," in *Audio Mostly 2024 - Explorations in Sonic Cultures*, Milan Italy: ACM, Sep. 18, 2024, pp. 260–269, ISBN: 979-8-4007-0968-5. DOI: 10.1145/3678299.3678324 Accessed: Feb. 18, 2026. [Online]. Available: https://dl.acm.org/doi/10.1145/3678299.3678324

[6] A. R. Vila, *Antoniorv6/SMT*, Feb. 14, 2026. Accessed: Feb. 19, 2026. [Online]. Available: https://github.com/antoniorv6/SMT

[7] "Ufal/olimpic-icdar24: Practical End-to-End Optical Music Recognition for Pianoform Music," Accessed: Feb. 18, 2026. [Online]. Available: https://github.com/ufal/olimpic-icdar24?tab=readme-ov-file

[8] Departament de Gestió, FIB–UPC, *Project management – module 2.4: Economic management*, GEP course material (internal). Covers budget structure (staff costs, general costs, amortisation, contingency, incidentals) and prescribes the x1.35 salary multiplier for Spanish employer social security contributions., Facultat d'Informàtica de Barcelona (FIB), Universitat Politècnica de Catalunya, 2025.

[9] Instituto Nacional de Estadística (INE). "Encuesta anual de estructura salarial (EAES). año 2023. datos definitivos." Official Spanish annual earnings survey (INEbase). The national mean gross annual salary in 2023 was 28,049.94 EUR. ICT-sector workers (NACE Section J) earn substantially above the mean, consistent with the 30,000–38,000 EUR gross range used here., Accessed: Mar. 9, 2026. [Online]. Available: https://www.ine.es/dyngs/INEbase/es/operacion.htm?c=Estadistica_C&cid=1254736177025&menu=ultiDatos&idp=1254735976596

[10] Ministerio de Inclusión, Seguridad Social y Migraciones. "Cotización y recaudación de trabajadores. regímenes." Official employer social security contribution rates for the Spanish General Regime. Total employer overhead is approximately 30 percent, commonly rounded to a x1.35 multiplier on gross salary in academic project budgets, as prescribed by the GEP economic management module., Accessed: Mar. 9, 2026. [Online]. Available: https://www.seg-social.es/wps/portal/wss/internet/Trabajadores/CotizacionRecaudacionTrabajadores/10721

[11] Departament de Gestió, FIB–UPC, *Sustainability analysis for the bachelor's thesis*, GEP course material (internal). Defines the sustainability matrix (Environmental, Economic, Social x PPP, Exploitation, Risks) and the questions for the initial and final milestones., Facultat d'Informàtica de Barcelona (FIB), Universitat Politècnica de Catalunya, 2018.

[12] Red Eléctrica de España (REE) / Redeia, "Informe del sistema eléctrico español 2023," Red Eléctrica de España, Tech. Rep., 2024, Annual report of the Spanish electricity system. The CO2 emission intensity of the Spanish grid in 2023 was approximately 0.23 kg CO2/kWh, as also reported by MITECO. Accessed: Mar. 9, 2026. [Online]. Available: https://www.ree.es/es/datos/publicaciones/informe-anual-sistema

[13] Ministerio para la Transición Ecológica y el Reto Demográfico (MITECO). "Factores de emisión: Electricidad." MITECO publishes annual CO2 emission factors for the Spanish electricity mix. The 2023 factor is approximately 0.23 kg CO2/kWh., Accessed: Mar. 9, 2026. [Online]. Available: https://www.miteco.gob.es/content/dam/miteco/es/energia/files-1/Eficiencia/RITE/documentosreconocidosrite/Otros%20documentos/Factores_emision_CO2.pdf